Dr. Lorenz Ennio Gärtner

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Technical Expertise

Statistics & ML	**Bayesian and frequentist inference**, MCMC posterior sampling (PyMC), profile likelihoods, constrained optimisation, uncertainty quantification, supervised learning, automatic differentiation
Programming	Python (**NumPy, SciPy, JAX, PyTorch**), C++, basic Julia, LaTeX, Mathematica
Tools	Git, Docker, CI/CD (GitHub Actions), pytest, packaging (PyPI, conda), HPC, profiling & optimisation, **open-source maintenance**, agentic development workflows
Languages	German (native), English (fluent); French, Italian (basic)

Work & Research Experience

Postdoctoral Researcher – *Democratizing statistical models*
2026 – present
BMFTR ErUM-Data project — *Ludwig Maximilians University, Munich, Germany*

- **Composable models:** Statistical methods and open-source tooling for probabilistic models that can be shared and combined across domains, with propagated uncertainties.
- **Verifiability:** Results from one experiment can be checked and reused by another, without re-running the original analysis.

Scientific Advisor to MP Lars Rohwer
Mar. – May 2026
German Parliament (Bundestag) — *Berlin, Germany*

- **Technical consulting:** Advised a Member of the German Bundestag on energy, economic policy, and emerging technologies, turning ambiguous topics into clear recommendations.
- **Communication & coordination:** Translated frontier research into briefings, speeches, and media outputs while aligning political and scientific stakeholders.

PhD Researcher – *Accelerating Discovery in Particle Physics*
2022 – 2025
Belle II experiment — *Ludwig Maximilians University, Munich, Germany*

- **Method:** Developed a novel statistical method for reinterpreting published measurements [3] (publications).
- **Software:** Built the open-source Python package redist, now applied within the Belle II [1, 2] and LHCb [4] collaborations.
- **Open source:** Contributed to pyhf, cabinetry, and eos: declarative model specification, inference, and theory predictions.
- **Strategy:** Input to the European Strategy for Particle Physics Update 2026, via the Forum for Open Science [5].
- **Talks:** 10+ international conference talks (ICHEP, PyHEP, WLCG/HSF, . . .).
- **Teaching:** Lectured on Bayesian and frequentist inference at the Inverted CERN School of Computing. Designed a scientific-Python course for 60+ students; tutorials for 50+ researchers.

Applied Researcher & Policy Coordinator
2021 – 2022
International Atomic Energy Agency, Fast Reactors Team, Internship — *Vienna, Austria*

- **Project coordination:** Coordinated a cross-disciplinary IAEA project connecting fusion and fission experts, organising workshops for 50+ experts.
- **Technical writing:** Authored and reviewed scientific proposals, papers, and presentations.

Earlier Research Positions: Machine Learning, Simulation & Scientific Software 2017 – 2022
Belle II; Max Planck Institute for extraterrestrial Physics; SLAC/Stanford *Germany; USA*

- **Applied machine learning:** Improved Belle II particle classification accuracy by >10% with a neural network replacing expensive detector simulations.
- **Large-scale simulation:** Evaluated observational astrophysics data against theoretical models using C++ Athena++ simulations.
- **Scientific tooling:** Extended the redMaPPer galaxy cluster finder with uncertainty estimation and masking corrections.

Education

PhD Physics: Particle Physics 2022 – 2025
summa cum laude (best possible) *Ludwig Maximilians University, Munich, Germany*
Accelerating discovery: model-agnostic likelihoods for the reinterpretation of particle physics results

MSc Physics: Theoretical & Computational Physics 2018 – 2020
Very good, 1.09 *Ludwig Maximilians University, Munich, Germany*

BSc Physics with Theoretical Physics 2015 – 2018
First Class, 86.7% (1st in cohort) *University of Manchester, Manchester, United Kingdom*

Professional Development 2024 – 2026

- BlueDot AGI Strategy Course (2026): technical trends of AI progress, “defence in depth” frameworks for AI safety, and AI governance.
- Machine Learning Summer School (MLSS) (2026): deep learning, probabilistic modelling, trustworthy ML (MPI for Intelligent Systems).
- Leading for Tomorrow (2025): DPG leadership program on project management, communication, and team building.
- CERN School of Computing (2024): 50+ hours on software development, machine learning, security and data management.
- Fast and Efficient Python Programming School (2024): code acceleration, profiling, and GPU programming.

Responsibility, Funding & Awards

- **ErUM-Data Project: DEMOS – 300 000 €** (2025): main author of the successful proposal (Federal Ministry of Research, Technology and Space).
- **Belle II Early Career Researchers Representative** (2024 – 2025): elected by peers, 1000+ member collaboration.
- **Climate Fresk Facilitator** (2022 – present): volunteer facilitator for IPCC-based climate workshops.
- **Franz Mandl Prize** (June 2018): best BSc theoretical physics graduate, University of Manchester.

Selected Publications

[1] **Lorenz Gärtner**, Nikolai Krug, Thomas Kuhr, Michael A. Schmidt, et al. *Constraints on invisible $B^+ \to K^+X$ decays from the Belle II $B^+ \to K^+\nu\bar{\nu}$ measurement*. In: *Phys. Rev. D* 114.3 (2026), p. 032003.

[2] Merna Abumusabh et al. *Model-agnostic likelihood for the reinterpretation of the $B^+ \to K^+\nu\bar{\nu}$ measurement at Belle II*. In: *Phys. Rev. D* 112.9 (2025), p. 092016.

[3] **Lorenz Gärtner**, Nikolai Hartmann, Lukas Heinrich, Malin Horstmann, et al. *Constructing model-agnostic likelihoods, a method for the reinterpretation of particle physics results*. In: *Eur. Phys. J. C* 84.7 (2024), p. 693.

[4] Johannes Albrecht, Florian Bernlochner, Marco Colonna, **Lorenz Gärtner**, et al. *Joint LHCb–Belle II prospects to constrain new physics in $B \to D^{(*)}\tau\nu$*. In: *Eur. Phys. J. C* 86.6 (2026), p. 718.

[5] Jonathan Butterworth et al. *Reinterpretation and preservation of data and analyses in HEP*. Community report. LHC Reinterpretation Forum, CERN-LPCC-2025-002, Mar. 2025.

[2] and [5] carry collaboration-wide author lists; [4] is alphabetical.